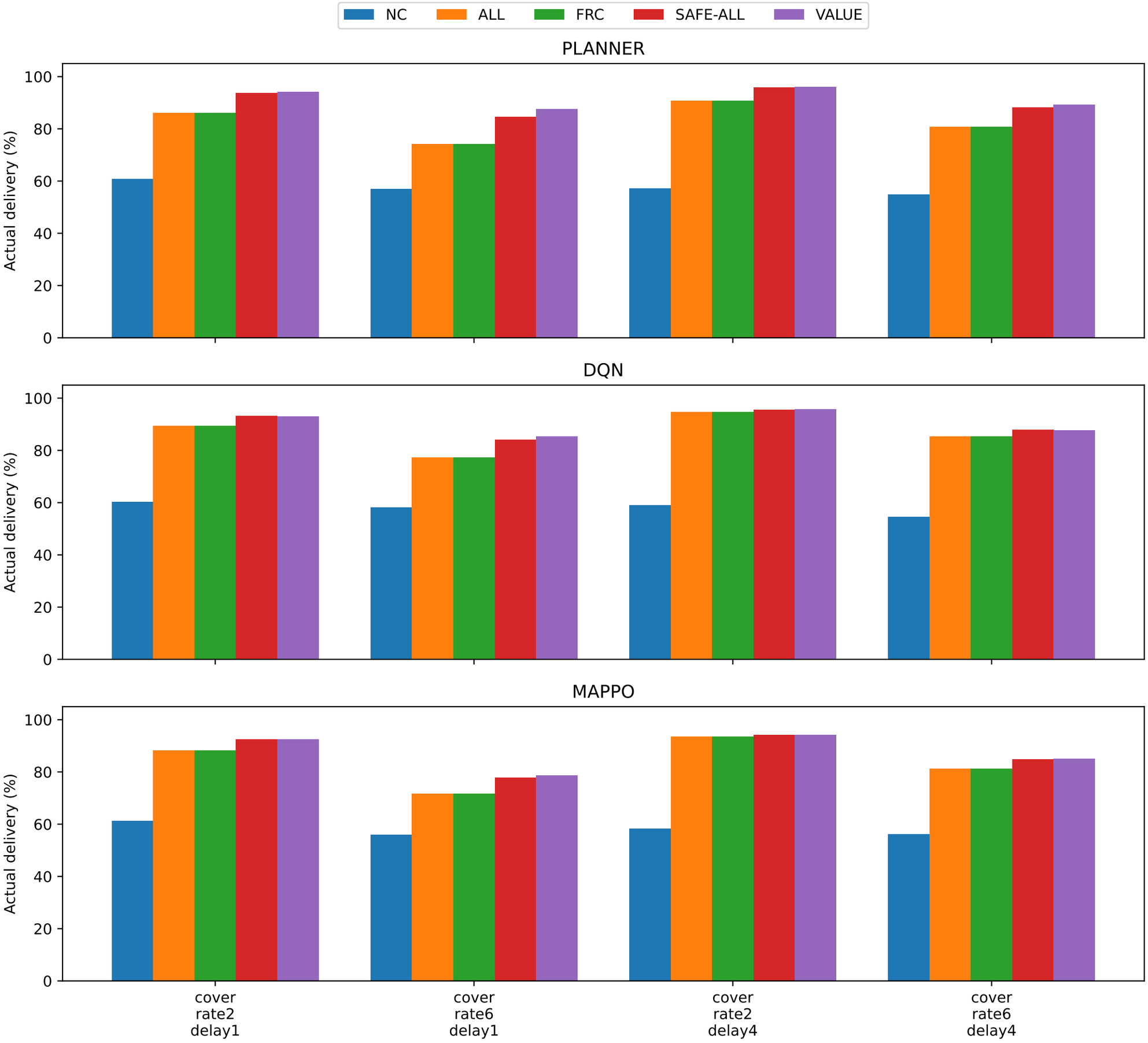
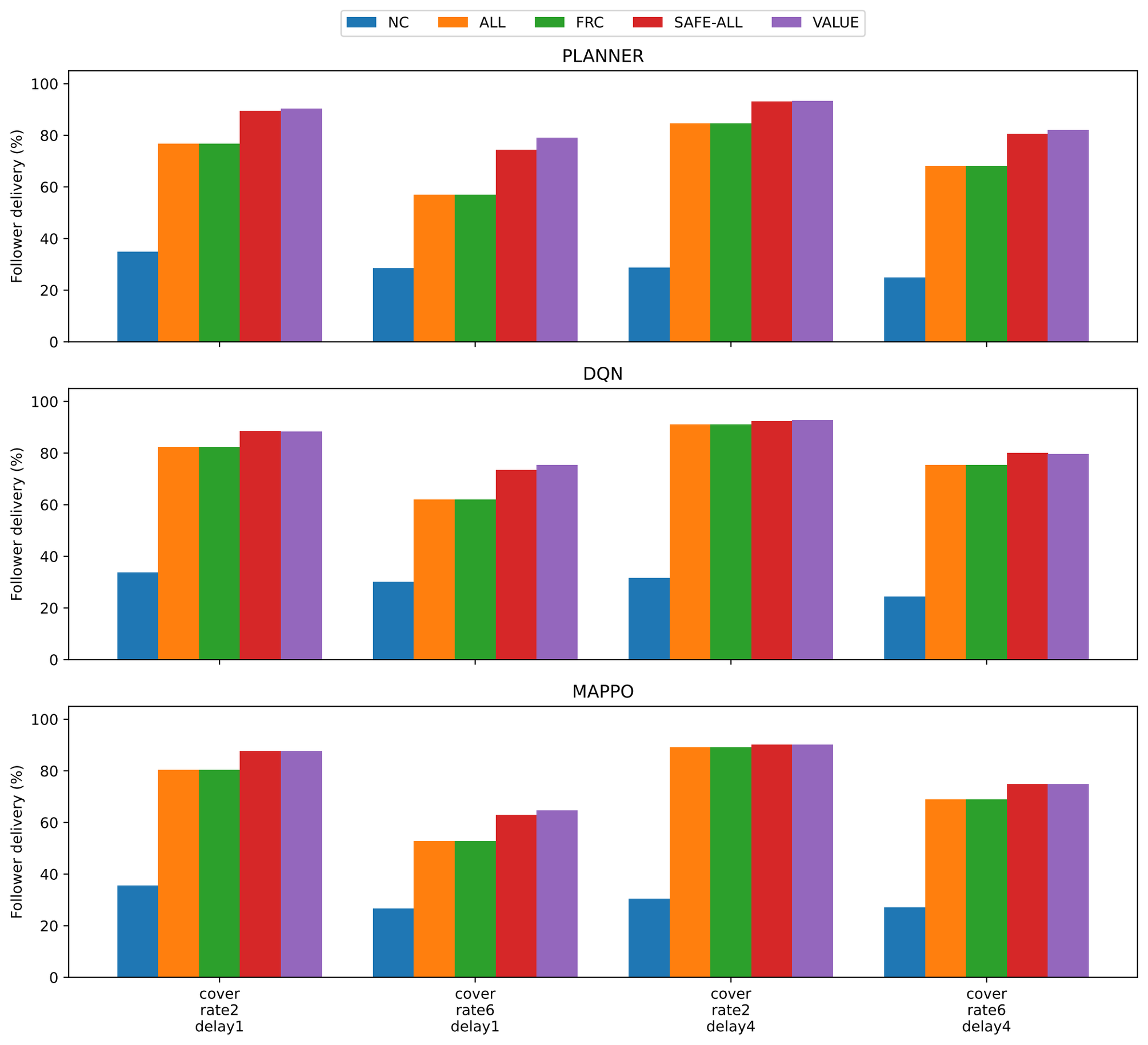
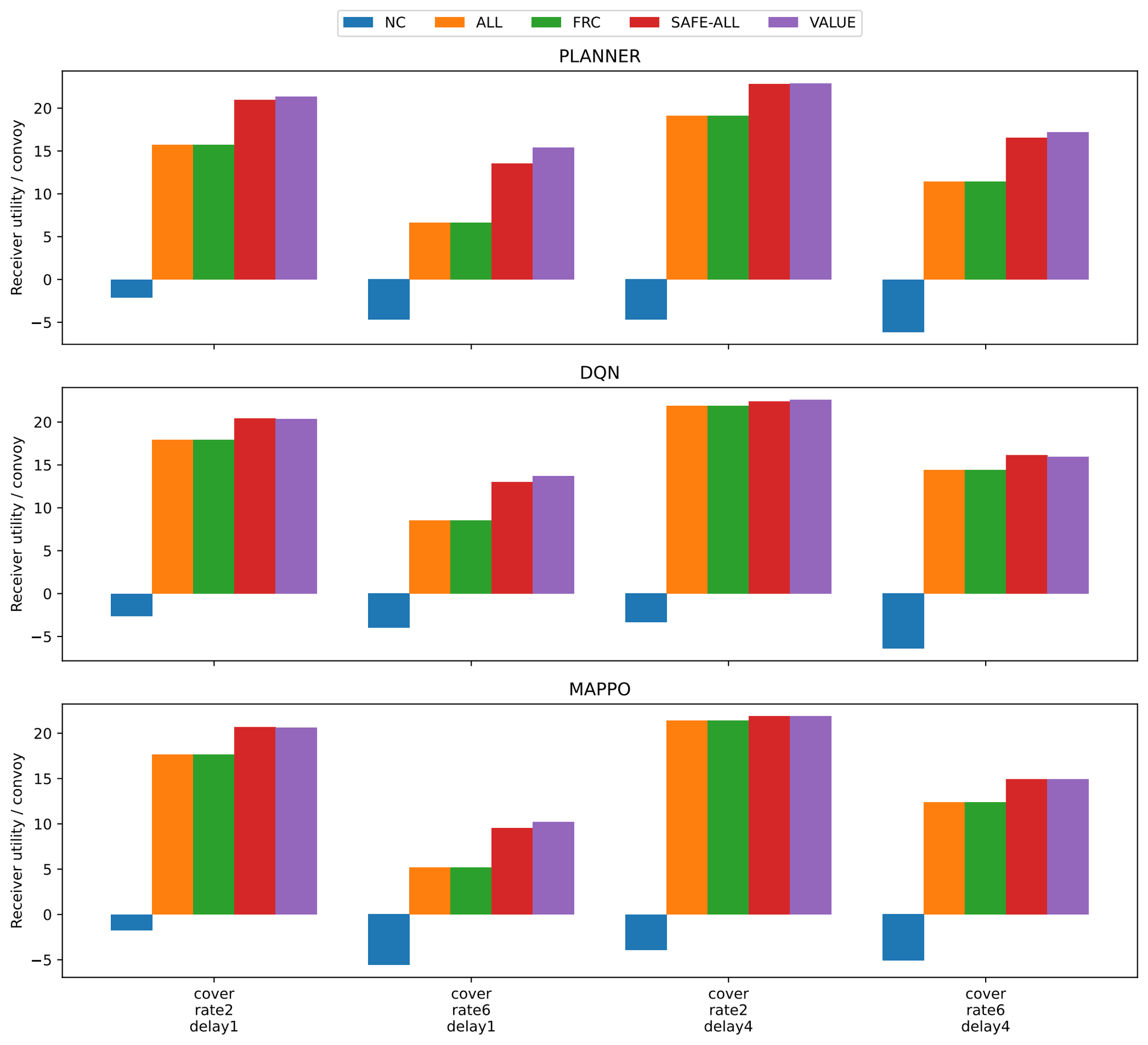
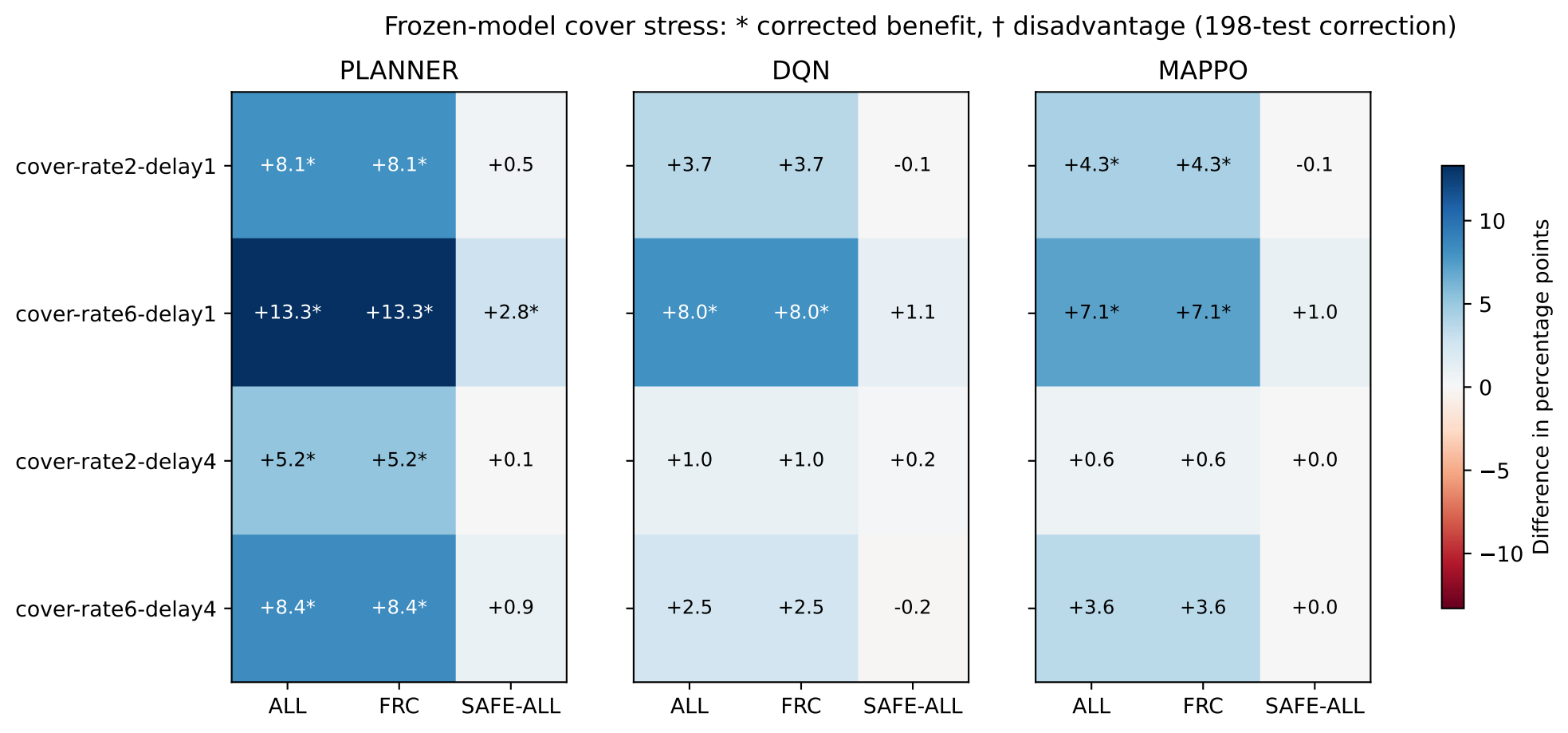

NC
ALL
FRC
SAFE-ALL
VALUE
PLANNER
DQN
MAPPO
Actual delivery (%)
0
20
40
60
80
100
cover rate2 delay1
cover rate6 delay1
cover rate2 delay4
cover rate6 delay4


NC
ALL
FRC
SAFE-ALL
VALUE
PLANNER
DQN
MAPPO
Follower delivery (%)
0
20
40
60
80
100
cover rate2 delay1
cover rate6 delay1
cover rate2 delay4
cover rate6 delay4


NC
ALL
FRC
SAFE-ALL
VALUE
PLANNER
DQN
MAPPO
Receiver utility / convoy
20
15
10
5
0
−5
cover rate2 delay1
cover rate6 delay1
cover rate2 delay4
cover rate6 delay4


Frozen-model cover stress: * corrected benefit, † disadvantage (198-test correction)
PLANNER
DQN
MAPPO
cover-rate2-delay1
cover-rate6-delay1
cover-rate2-delay4
cover-rate6-delay4
+8.1*
+8.1*
+0.5
+3.7
+3.7
-0.1
+4.3*
+4.3*
-0.1
+13.3*
+13.3*
+2.8*
+8.0*
+8.0*
+1.1
+7.1*
+7.1*
+1.0
+5.2*
+5.2*
+0.1
+1.0
+1.0
+0.2
+0.6
+0.6
+0.0
+8.4*
+8.4*
+0.9
+2.5
+2.5
-0.2
+3.6
+3.6
+0.0
ALL
FRC
SAFE-ALL
ALL
FRC
SAFE-ALL
ALL
FRC
SAFE-ALL
10
5
0
−5
−10
Difference in percentage points

NC ALL FRC SAFE-ALL VALUE

PLANNER

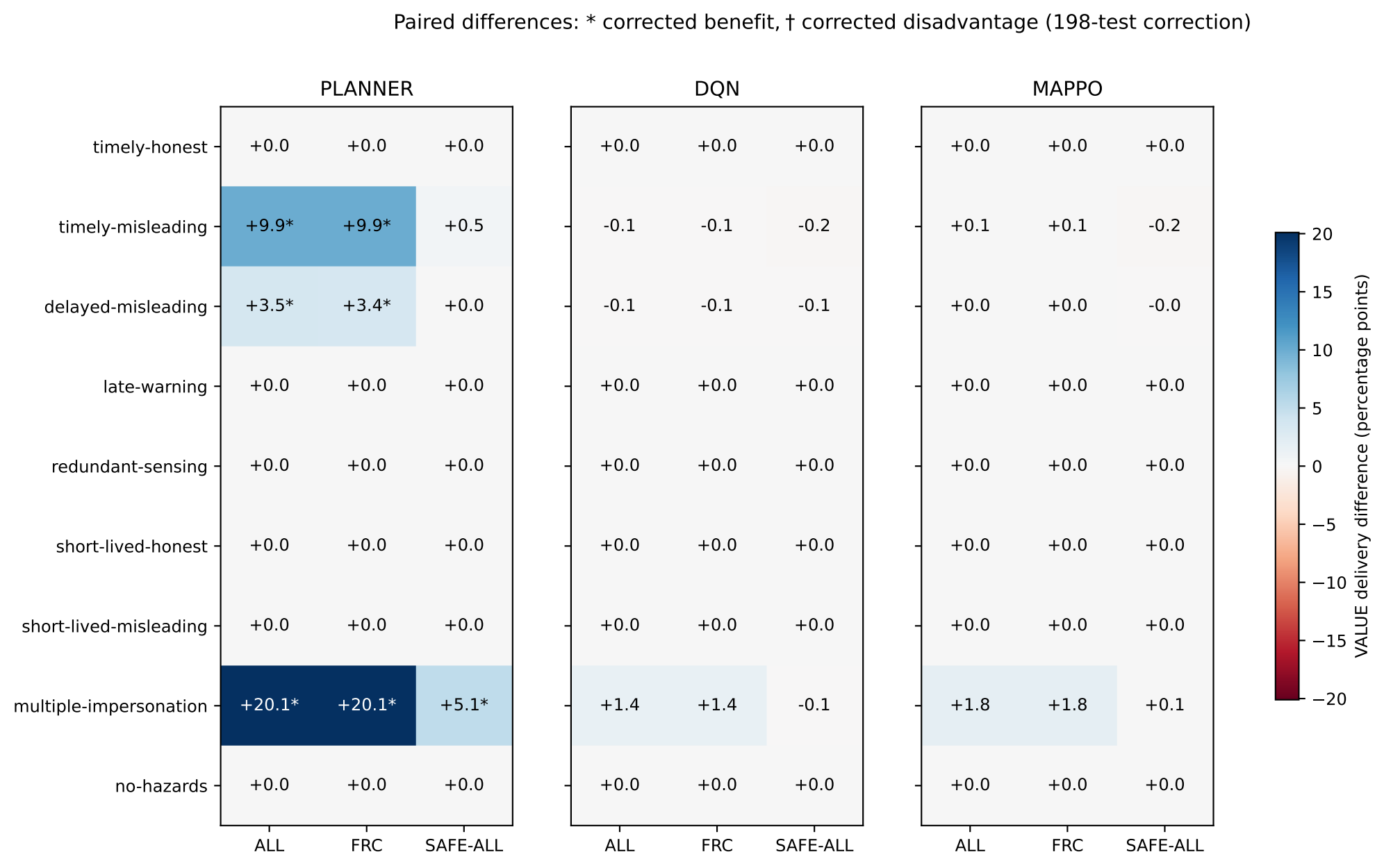
DQN

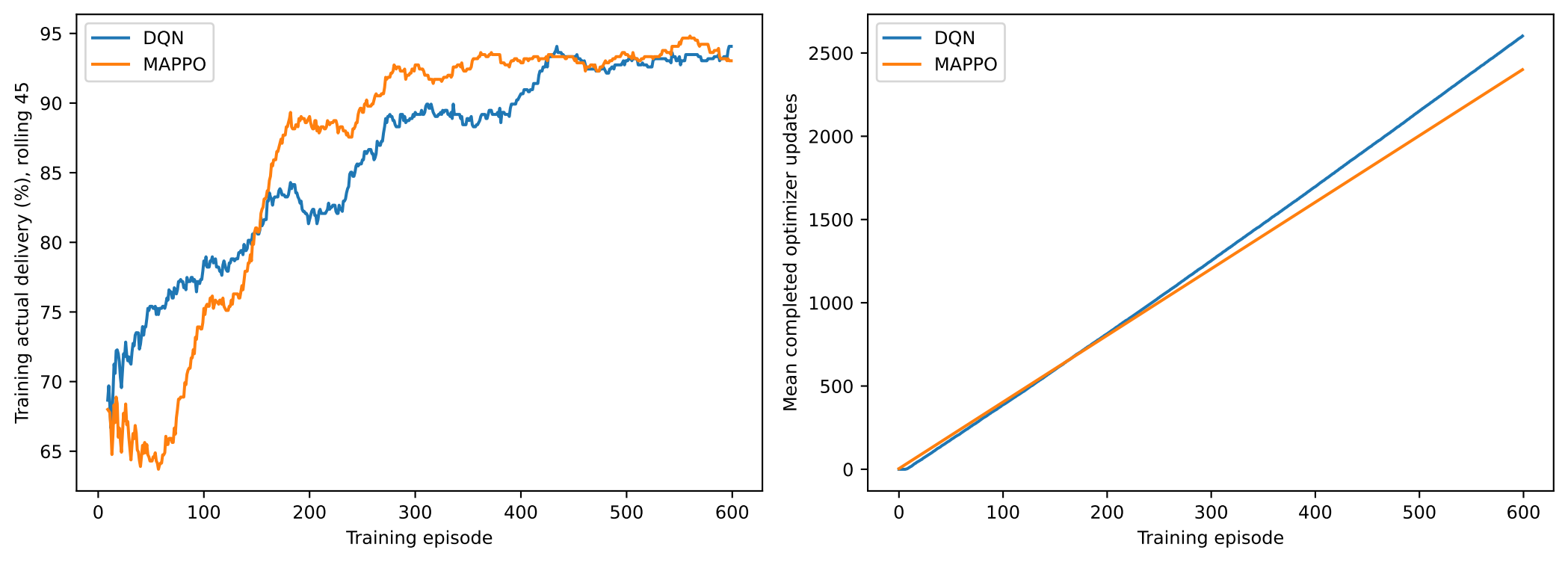
MAPPO

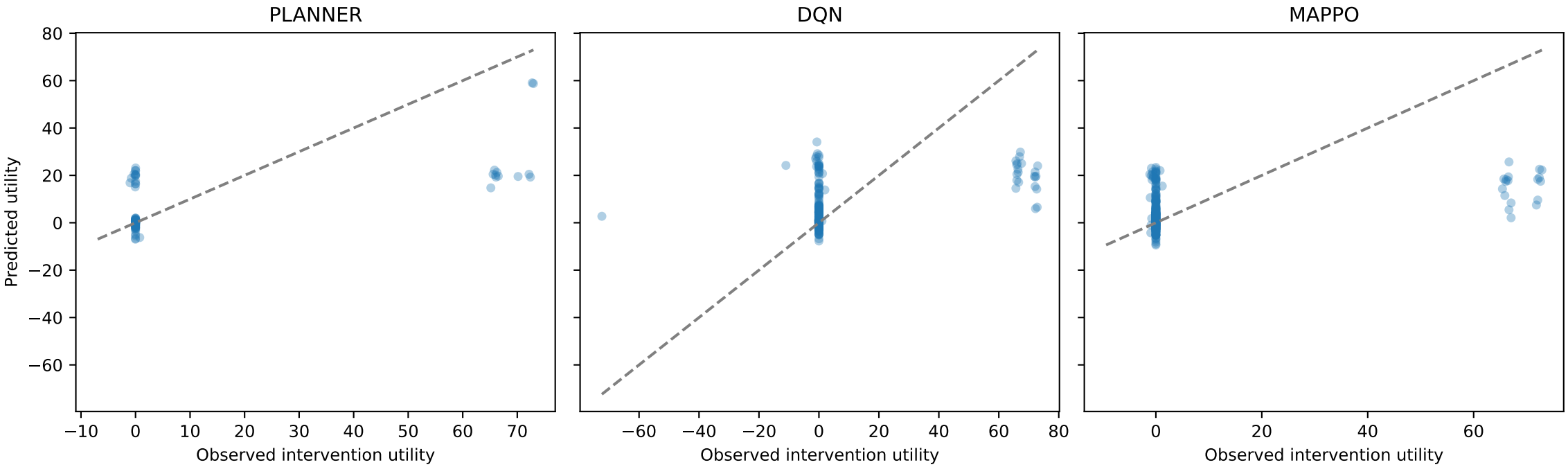
Actual delivery (%)

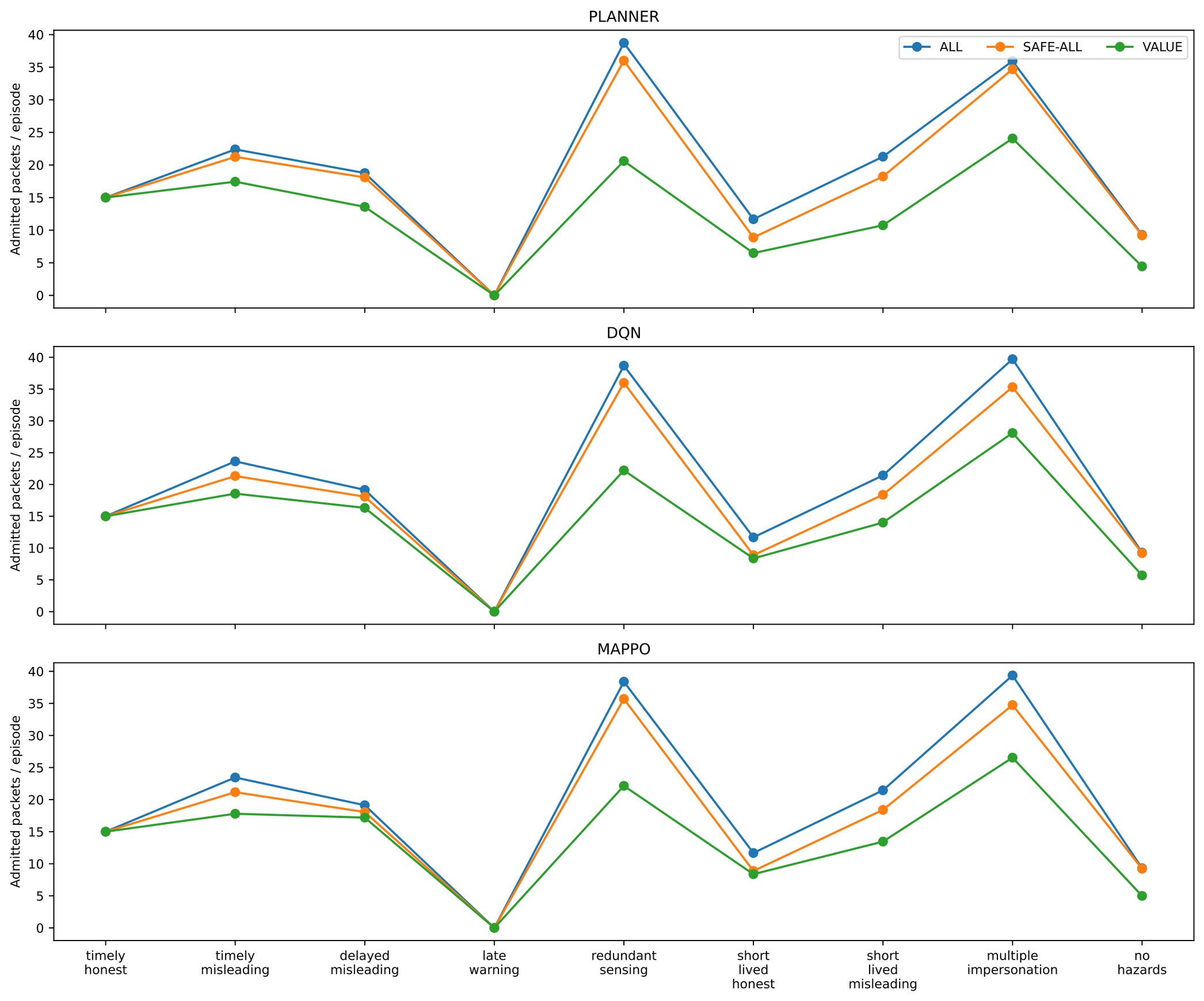
0 20 40 60 80 100

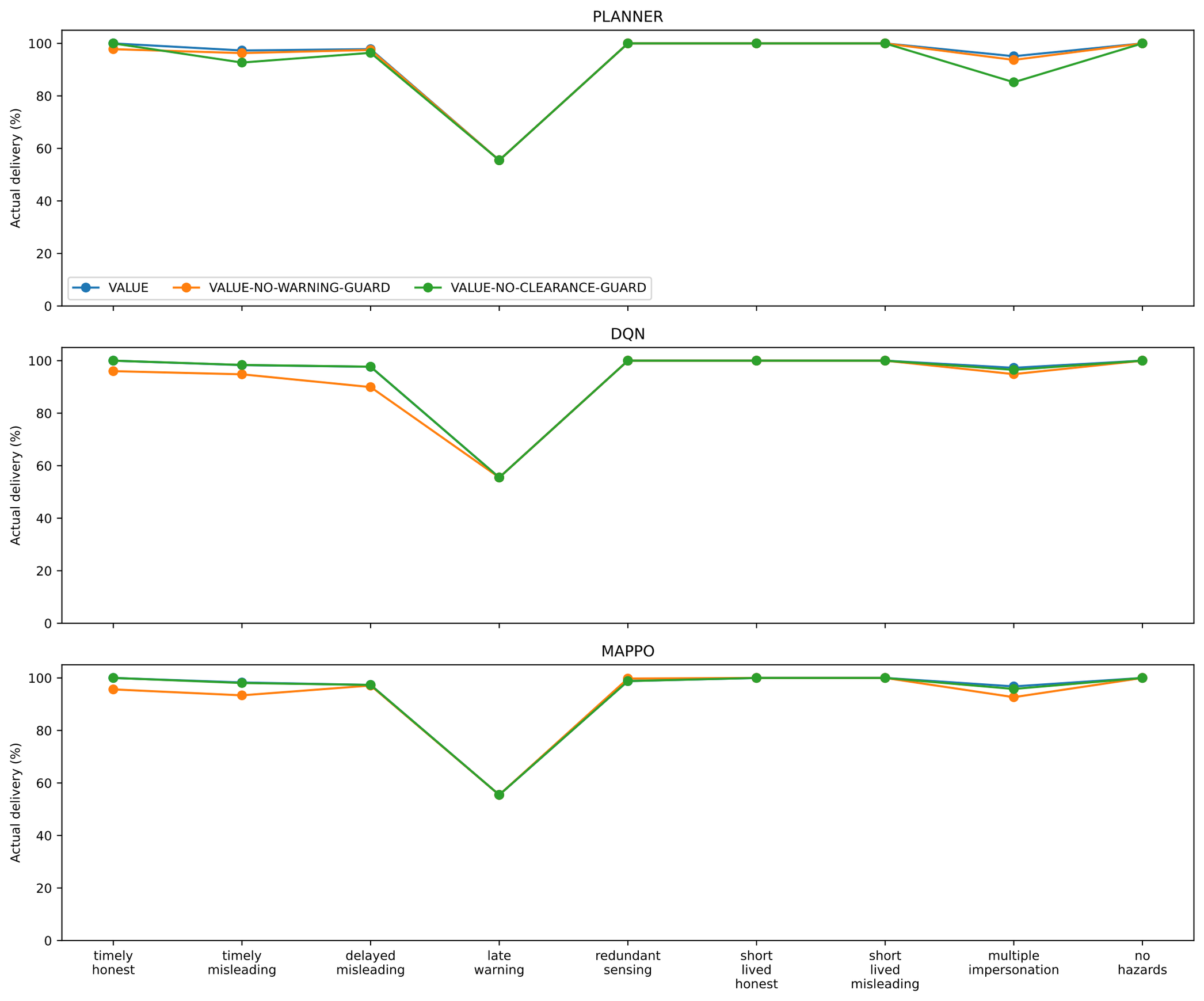
timely honest
timely misleading
delayed misleading
late warning
redundant sensing
short lived honest
short lived misleading
multiple impersonation
no hazards

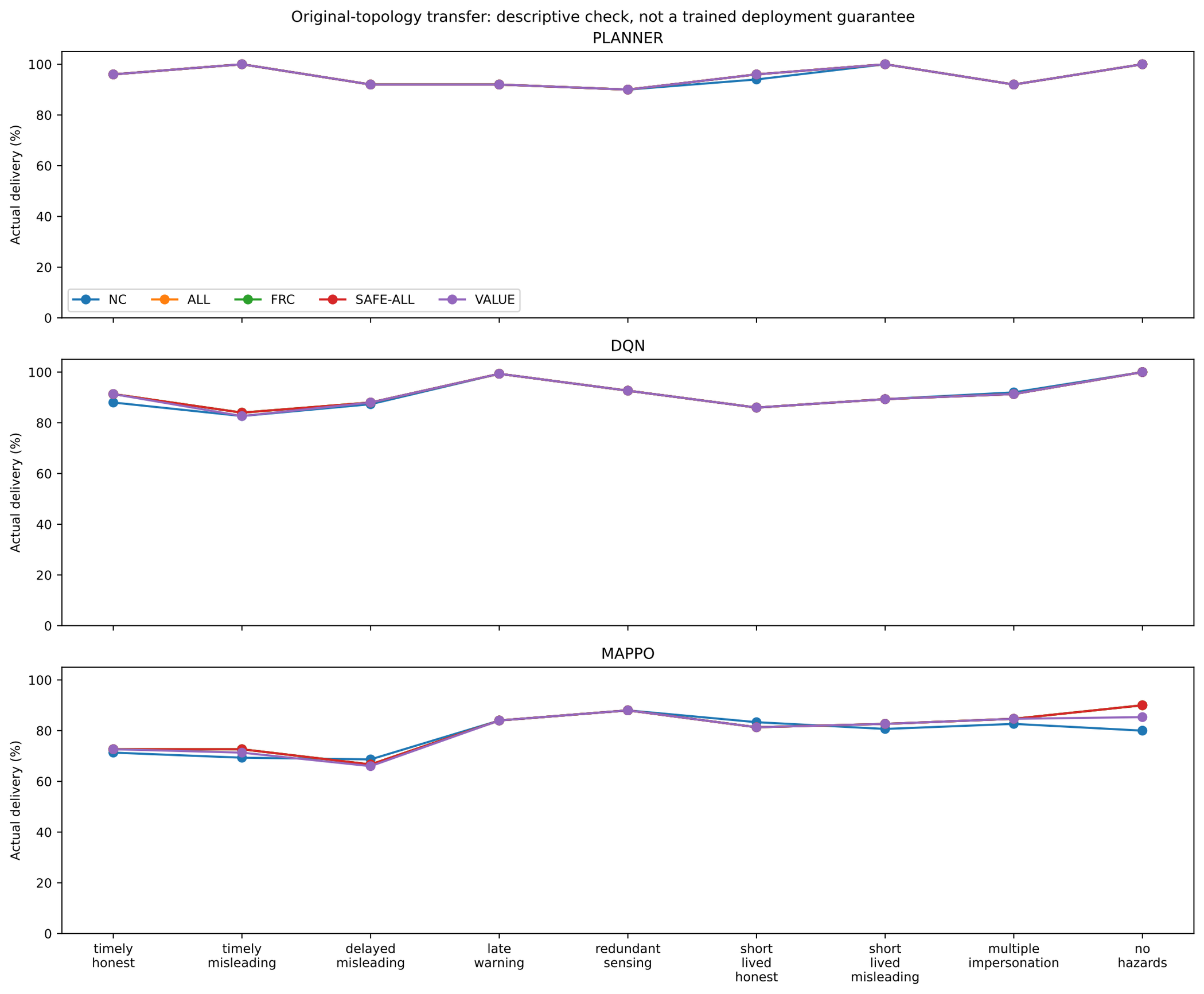

Paired differences: * corrected benefit, † corrected disadvantage (198-test correction)
PLANNER
DQN
MAPPO
timely-honest
timely-misleading
delayed-misleading
late-warning
redundant-sensing
short-lived-honest
short-lived-misleading
multiple-impersonation
no-hazards
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+9.9* +9.9* +0.5
-0.1 -0.1 -0.2
+0.1 +0.1 -0.2
+3.5* +3.4* +0.0
-0.1 -0.1 -0.1
+0.0 +0.0 -0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+20.1* +20.1* +5.1*
+1.4 +1.4 -0.1
+1.8 +1.8 +0.1
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
ALL FRC SAFE-ALL
ALL FRC SAFE-ALL
ALL FRC SAFE-ALL
20
15
10
5
0
−5
−10
−15
−20
VALUE delivery difference (percentage points)

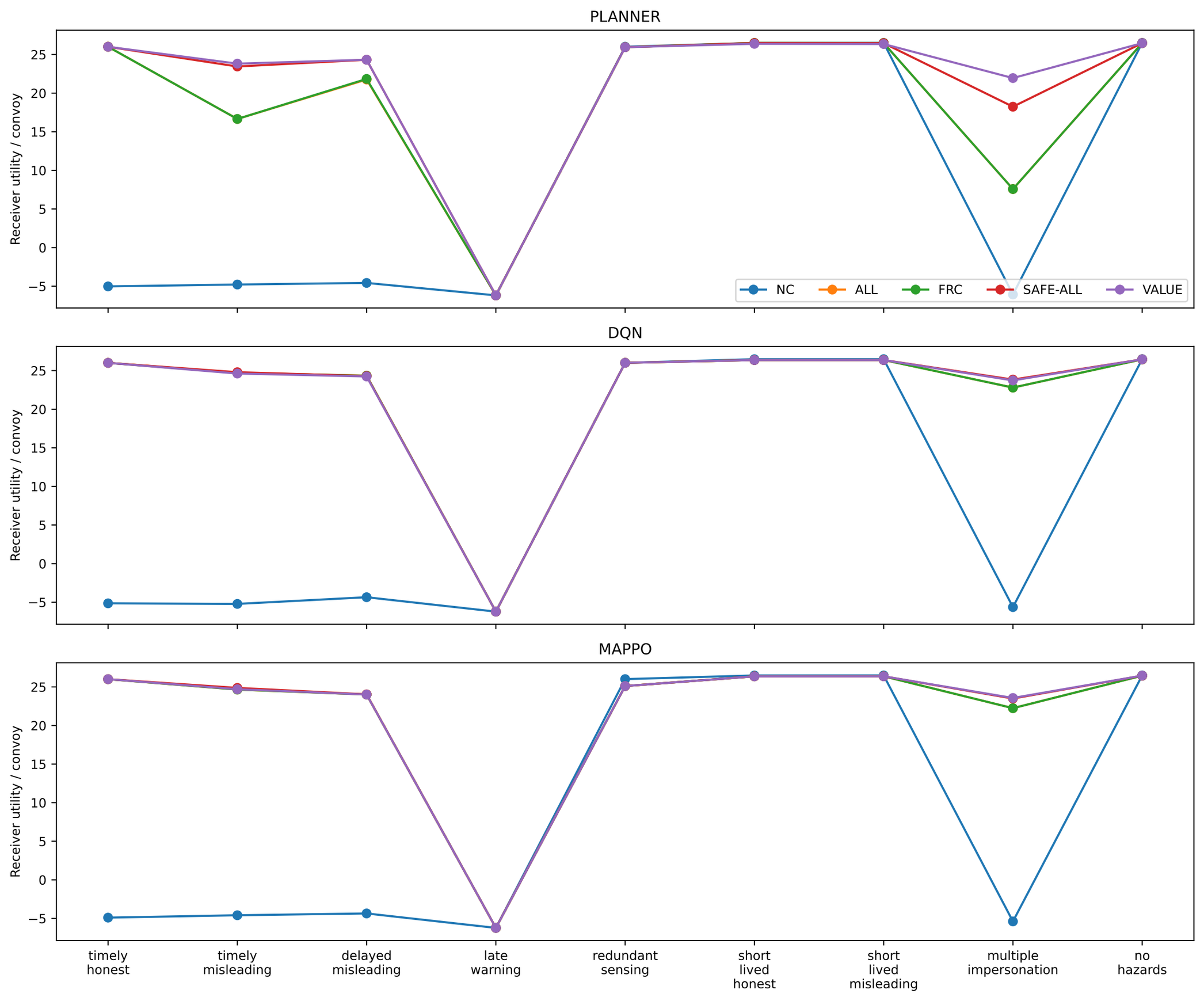

DQN
MAPPO
Training actual delivery (%), rolling 45
Training episode
Mean completed optimizer updates
Training episode


Internal held-out causal labels: episodes excluded from model fitting
PLANNER
DQN
MAPPO
Predicted utility
Observed intervention utility
Observed intervention utility
Observed intervention utility


PLANNER
DQN
MAPPO
ALL
SAFE-ALL
VALUE
Admitted packets / episode
0
5
10
15
20
25
30
35
40
timely honest
timely misleading
delayed misleading
late warning
redundant sensing
short lived honest
short lived misleading
multiple impersonation
no hazards


PLANNER
DQN
MAPPO
Actual delivery (%)
0
20
40
60
80
100
VALUE
VALUE-NO-WARNING-GUARD
VALUE-NO-CLEARANCE-GUARD
timely honest
timely misleading
delayed misleading
late warning
redundant sensing
short lived honest
short lived misleading
multiple impersonation
no hazards


Original-topology transfer: descriptive check, not a trained deployment guarantee
PLANNER
DQN
MAPPO
Actual delivery (%)
0
20
40
60
80
100
NC
ALL
FRC
SAFE-ALL
VALUE
timely honest
timely misleading
delayed misleading
late warning
redundant sensing
short lived honest
short lived misleading
multiple impersonation
no hazards


PLANNER
DQN
MAPPO
Receiver utility / convoy
NC
ALL
FRC
SAFE-ALL
VALUE
timely honest
timely misleading
delayed misleading
late warning
redundant sensing
short lived honest
short lived misleading
multiple impersonation
no hazards

NC ALL FRC SAFE-ALL VALUE

PLANNER

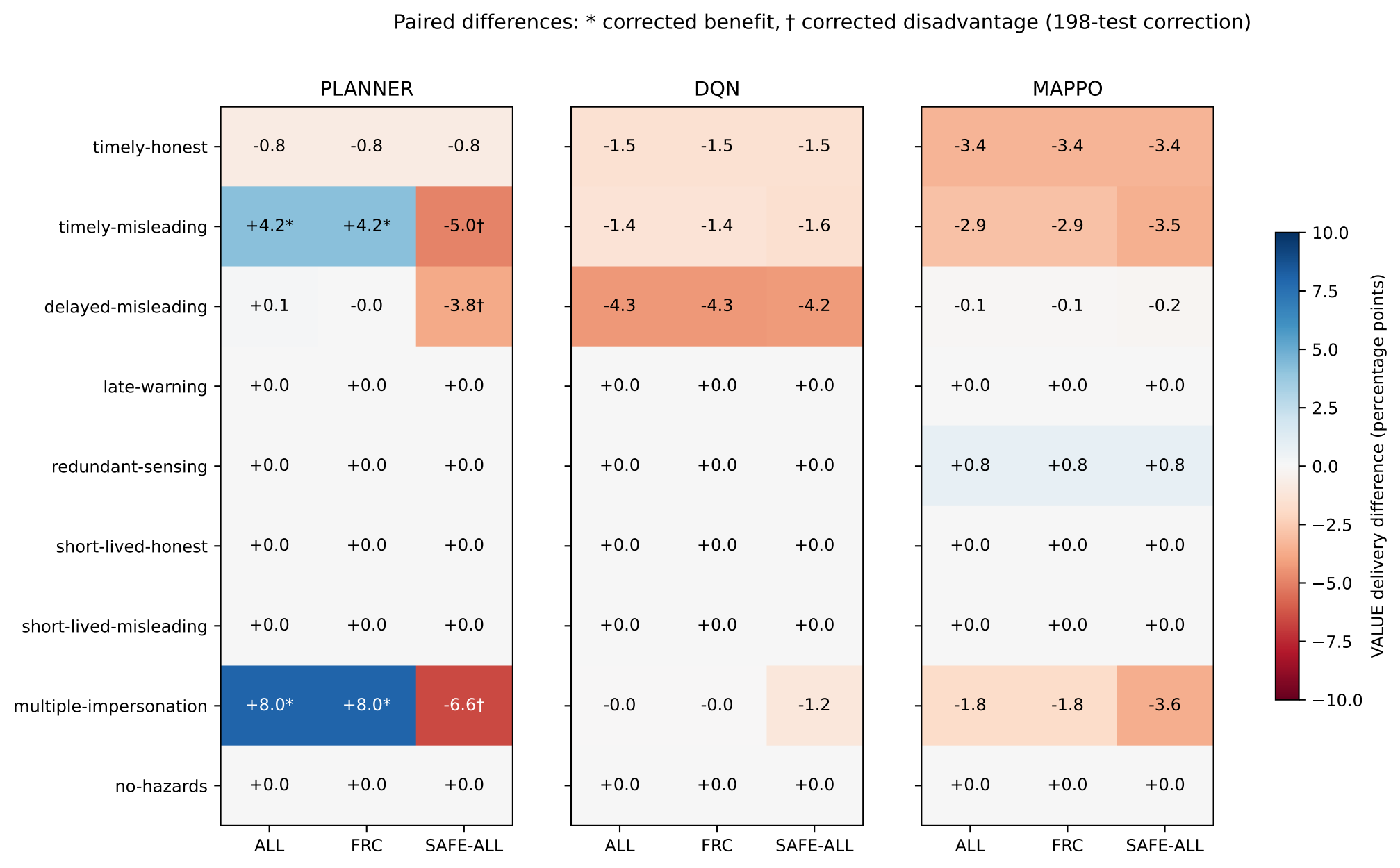
DQN

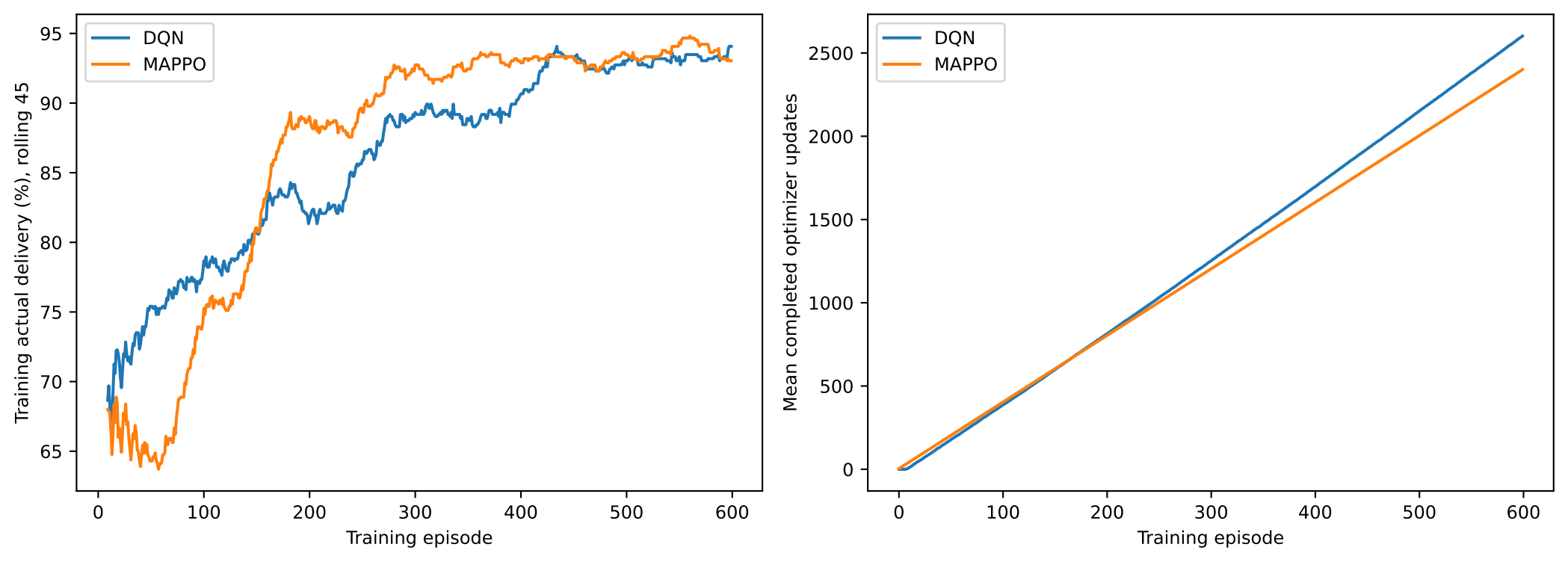
MAPPO

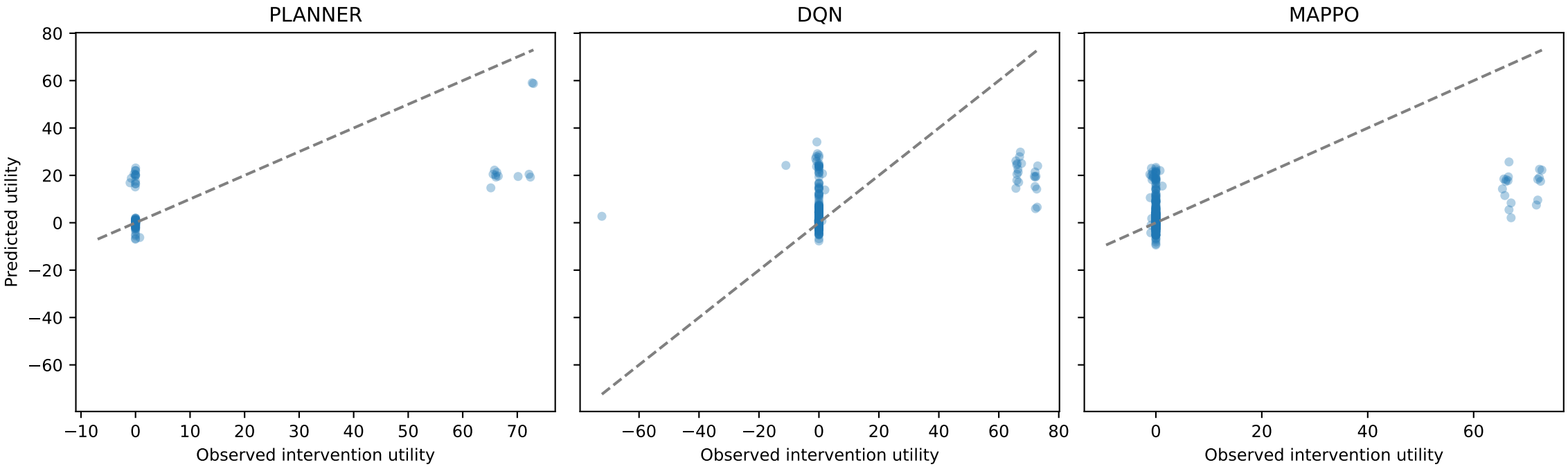
Actual delivery (%)

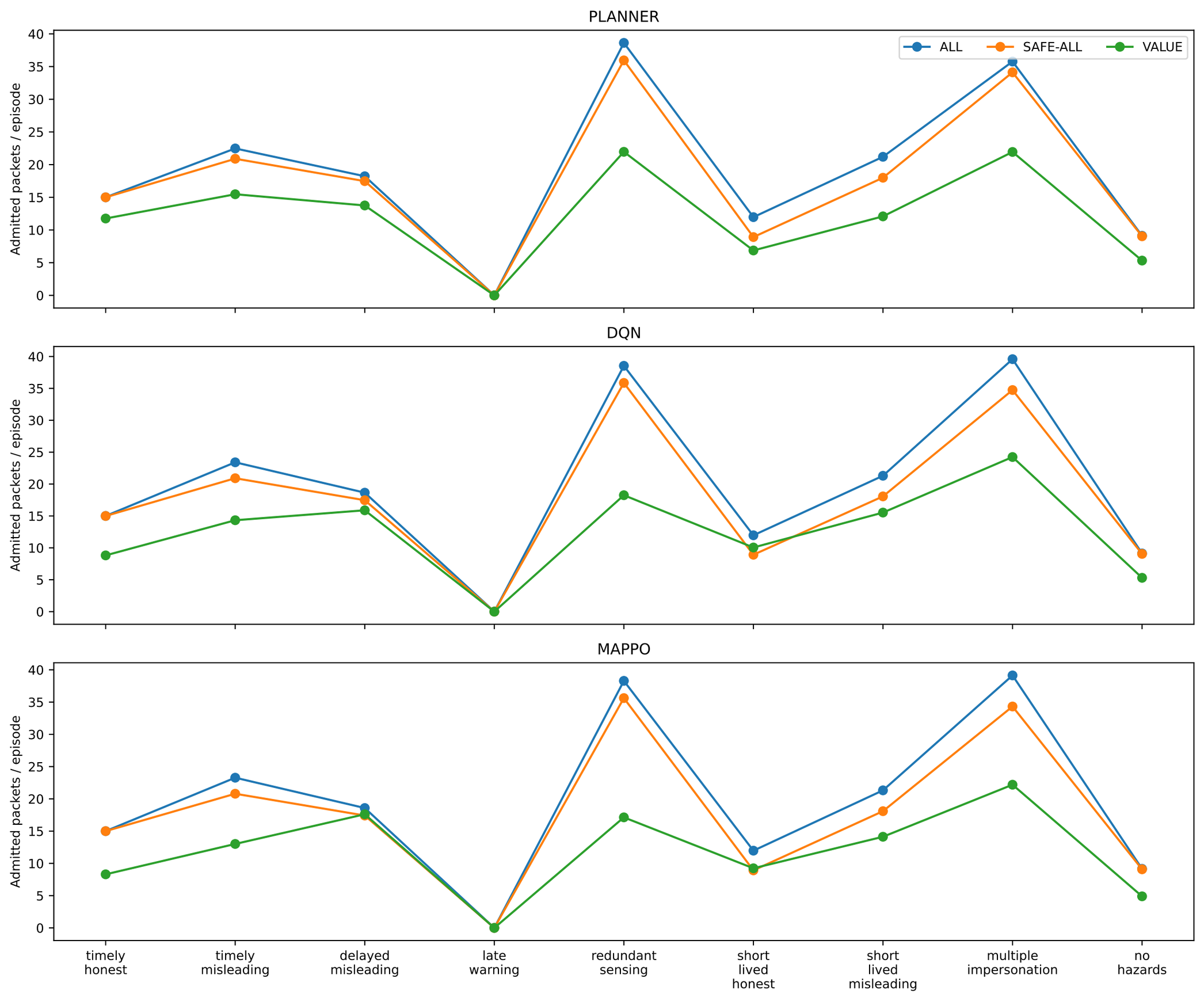
0 20 40 60 80 100

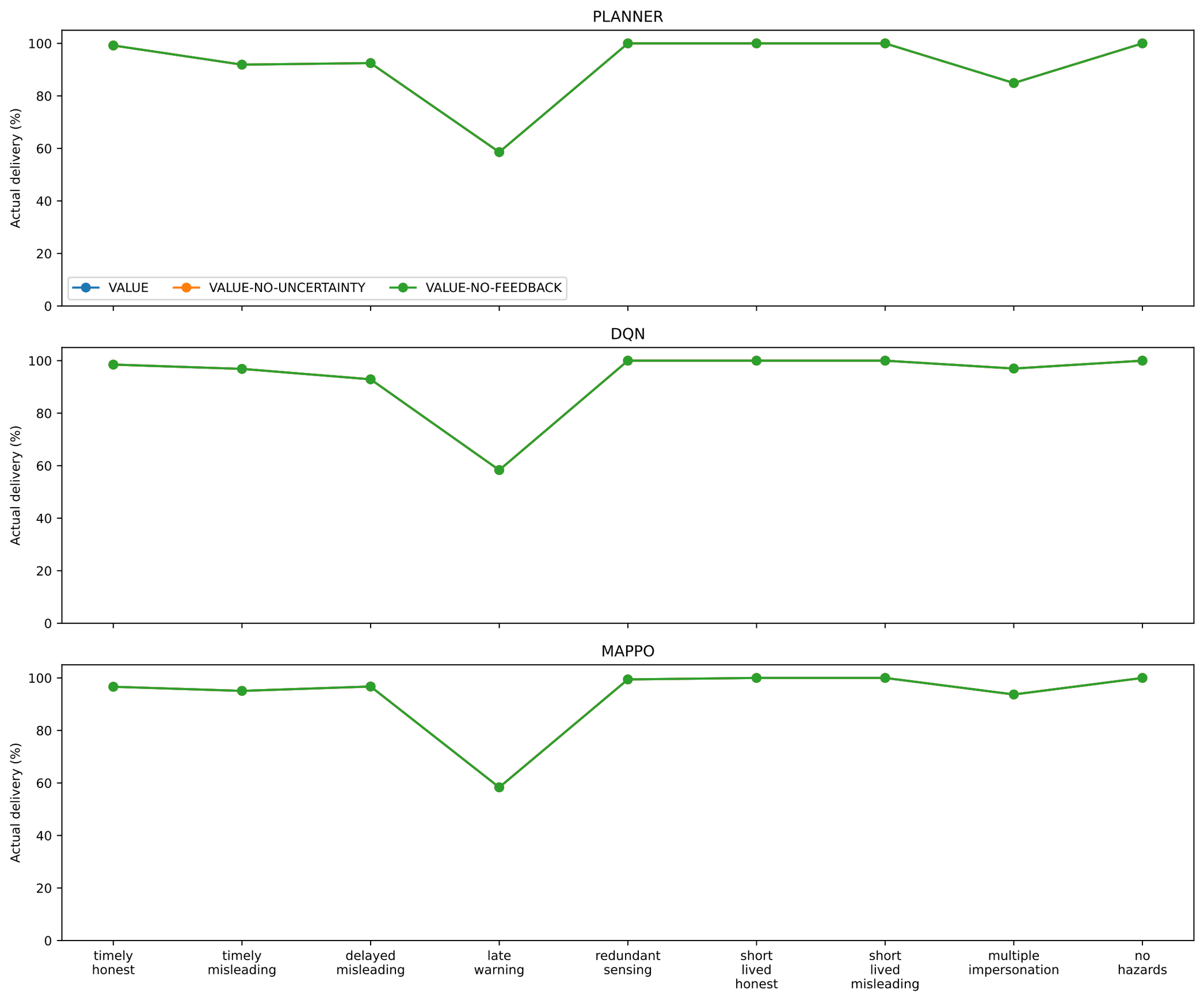
timely honest
timely misleading
delayed misleading
late warning
redundant sensing
short lived honest
short lived misleading
multiple impersonation
no hazards

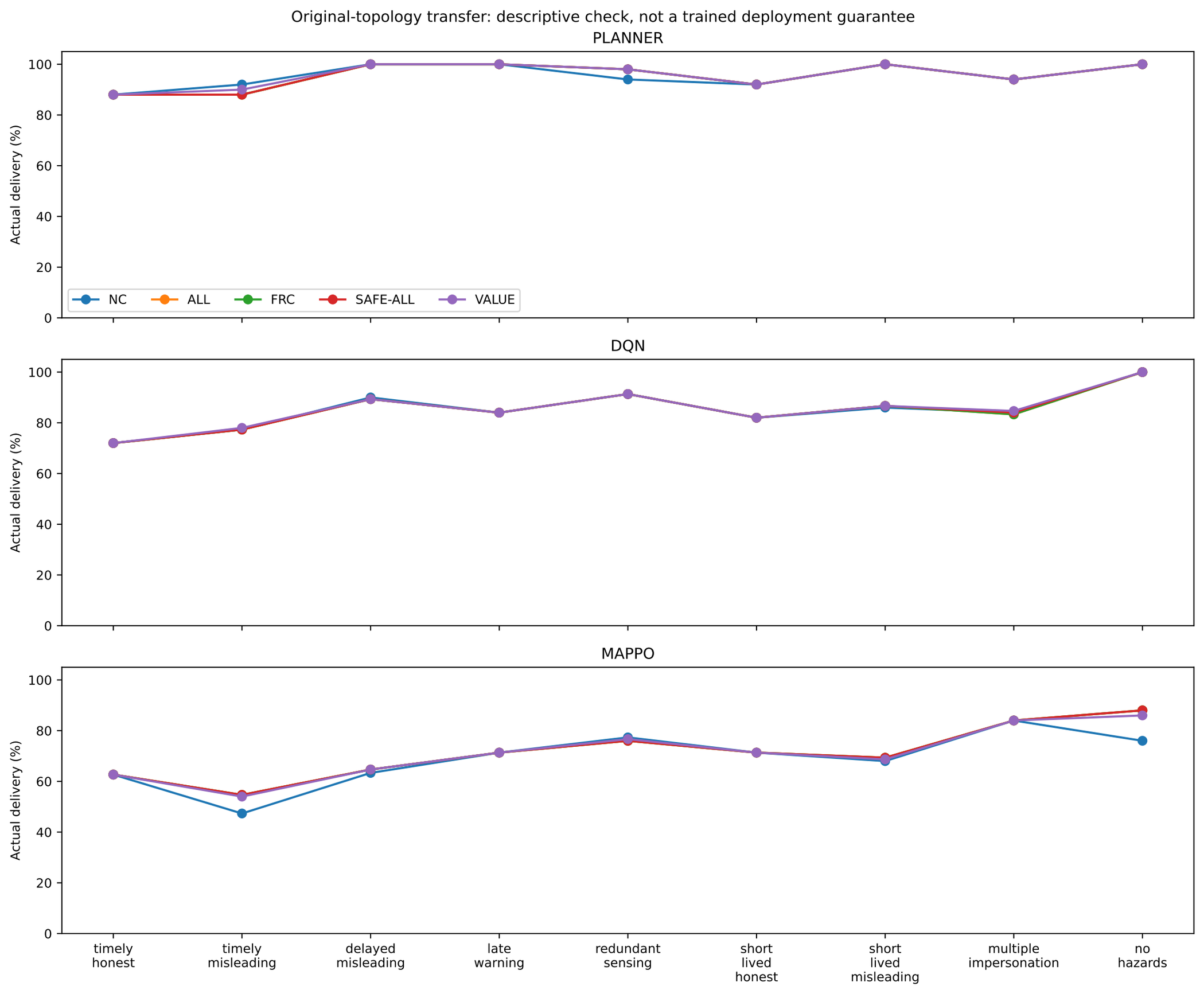

Paired differences: * corrected benefit, † corrected disadvantage (198-test correction)
PLANNER
DQN
MAPPO
timely-honest
timely-misleading
delayed-misleading
late-warning
redundant-sensing
short-lived-honest
short-lived-misleading
multiple-impersonation
no-hazards
-0.8 -0.8 -0.8
-1.5 -1.5 -1.5
-3.4 -3.4 -3.4
+4.2* +4.2* -5.0†
-1.4 -1.4 -1.6
-2.9 -2.9 -3.5
+0.1 -0.0 -3.8†
-4.3 -4.3 -4.2
-0.1 -0.1 -0.2
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.8 +0.8 +0.8
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+8.0* +8.0* -6.6†
-0.0 -0.0 -1.2
-1.8 -1.8 -3.6
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
ALL FRC SAFE-ALL
ALL FRC SAFE-ALL
ALL FRC SAFE-ALL
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
VALUE delivery difference (percentage points)

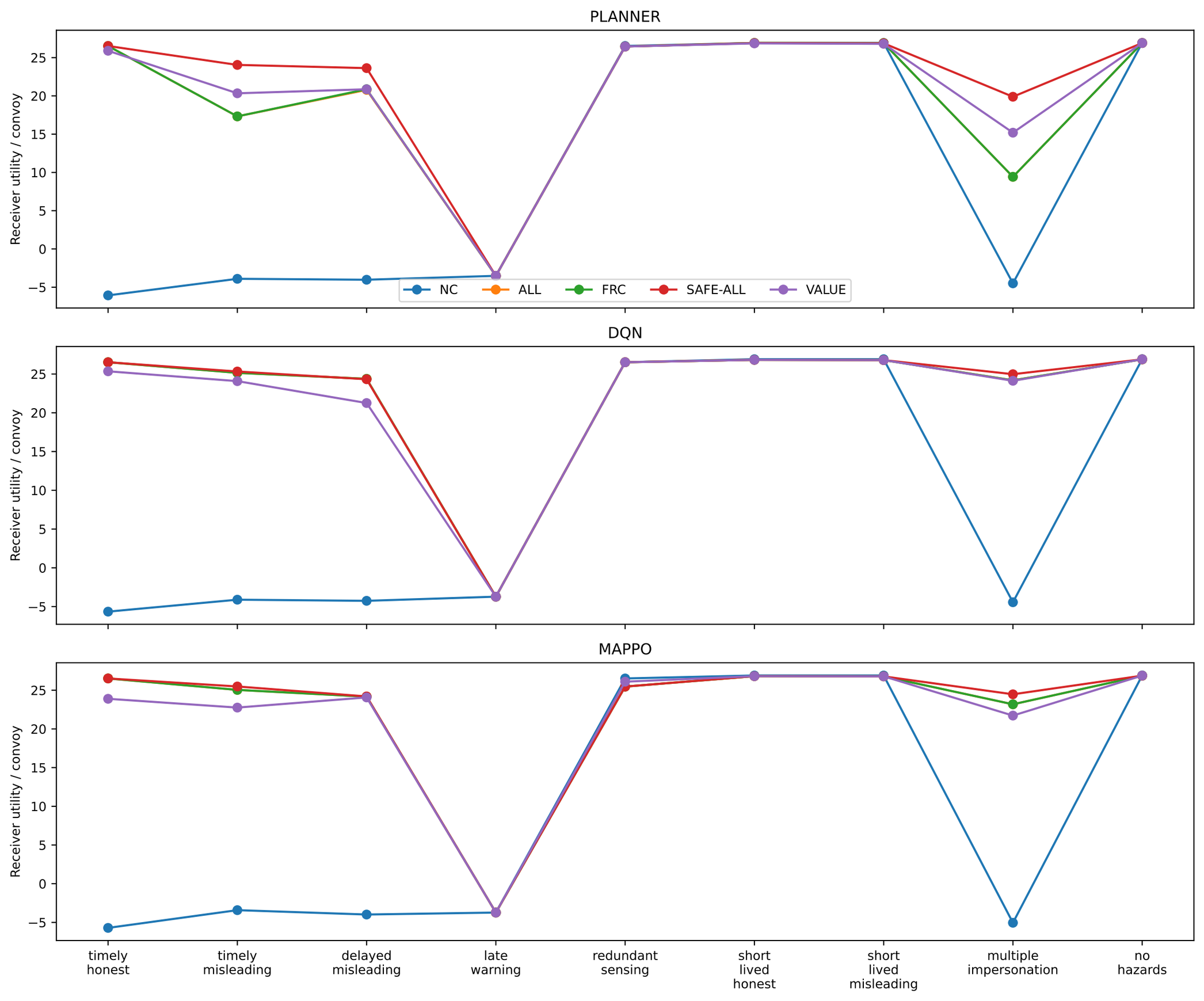

DQN
MAPPO
Training actual delivery (%), rolling 45
Training episode
Mean completed optimizer updates
Training episode


Internal held-out causal labels: episodes excluded from model fitting
PLANNER
DQN
MAPPO
Predicted utility
Observed intervention utility
Observed intervention utility
Observed intervention utility


PLANNER
DQN
MAPPO
ALL
SAFE-ALL
VALUE
Admitted packets / episode
0
5
10
15
20
25
30
35
40
timely honest
timely misleading
delayed misleading
late warning
redundant sensing
short lived honest
short lived misleading
multiple impersonation
no hazards


PLANNER
DQN
MAPPO
Actual delivery (%)
0
20
40
60
80
100
VALUE
VALUE-NO-UNCERTAINTY
VALUE-NO-FEEDBACK
timely honest
timely misleading
delayed misleading
late warning
redundant sensing
short lived honest
short lived misleading
multiple impersonation
no hazards


Original-topology transfer: descriptive check, not a trained deployment guarantee
PLANNER
DQN
MAPPO
Actual delivery (%)
0
20
40
60
80
100
NC
ALL
FRC
SAFE-ALL
VALUE
timely honest
timely misleading
delayed misleading
late warning
redundant sensing
short lived honest
short lived misleading
multiple impersonation
no hazards


PLANNER
DQN
MAPPO
Receiver utility / convoy
NC
ALL
FRC
SAFE-ALL
VALUE
timely honest
timely misleading
delayed misleading
late warning
redundant sensing
short lived honest
short lived misleading
multiple impersonation
no hazards